Instructions:

- **Due March 20, 11:59PM uploaded to Blackboard as a single pdf.**
- Each problem is worth $\dfrac{100}{\text{number of problems}}$ points.
- Start each problem on a new page and put a box around your final answer to help the grader.
- Label your file as `ENME403_S26_HW06_yourname.pdf`.
- If you use Simulink to generate results for the homework submission, include screenshots of the block diagrams and all relevant plots.
- **Use of AI Tools:** The use of AI software to prepare your submission is strictly prohibited. If we determine that AI generated a submission, the work will receive a grade of zero and will be reported to the Academic Conduct Committee. You may use AI tools to aid your understanding, but all submitted work must be entirely your own.
- Failure to follow these guidelines will result in zero points.

Learning Objective: This homework develops students' ability to analyze linear control systems using transfer functions and block diagrams. Students will assess internal and BIBO stability from pole locations, derive sensitivity and transmissibility functions for feedback systems, and relate pole locations to time-domain performance metrics such as rise time and settling time. Students will also use Simulink to simulate system responses and verify analytical predictions using the final value theorem.

Problem 1. In Simulink, implement

$$G(s) = \frac{s-2}{s^2+3s+2}. \tag{1}$$

1. Determine the poles and zeros of $G(s)$. Is the system asymptotically stable?.
2. For each of the following inputs, predict qualitatively whether the output will decay, remain bounded, or grow without bound.

   (a) $u(t) = \sin(t)$

   (b) $u(t) = e^{-2t}$

   (c) $u(t) = e^{2t}$

   Explain your reasoning based on the pole–zero structure of $G(s)$ and the form of the input signal.
3. Implement the system in Simulink and simulate the response for each input for $0 \le t \le 10$.
4. Plot the output for each case and compare the simulation results with your predictions.
5. Explain why the response to $u(t) = e^{2t}$ is qualitatively different from the responses to the other inputs.

Problem 2. Consider the open-loop interconnection shown in Figure 1.

1. Derive the following transfer functions

$$G_{yr}(s), \qquad G_{yu}(s), \qquad G_{ur}(s). \tag{2}$$

2. For each of the transfer functions above, determine whether it is BIBO stable. Explain your answer using the pole locations.
3. Is the interconnection internally stable? Explain your answer.

4. Implement the system shown in Figure 1 in Simulink. Let $r(t) = 1(t)$.
   Plot the control signal $u(t)$ and the output $y(t)$ for $0 \le t \le 10$.
5. Explain why the simulation may appear well-behaved even though the system is not internally stable.

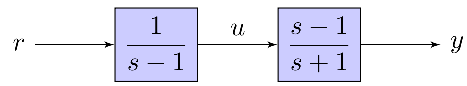

Figure 1: Open-loop control.

Hint:

- A transfer function is *BIBO stable* if every bounded input produces a bounded output. For linear time-invariant systems, this occurs if and only if all poles of the transfer function have strictly negative real parts.
- A system interconnection is *internally stable* if all internal signals remain bounded for bounded inputs and all internal modes are stable. Internal stability requires that *all* poles of the closed-loop interconnection (including those that may cancel in input–output transfer functions) lie in the open left-half plane.

Problem 3. For each of the feedback systems shown below, determine the *sensitivity* and *transmissibility* transfer functions

$$S(s) \triangleq G_{er}(s), \tag{3}$$

$$T(s) \triangleq G_{yr}(s), \tag{4}$$

where $r(t)$ is the reference command, $y(t)$ is the output, and $e(t) \triangleq r(t) - y(t)$ is the tracking error. For each system

1. Derive $S(s)$ and $T(s)$.
2. Determine whether the identity $S(s) + T(s) = 1$ holds.

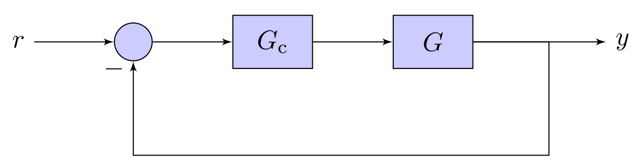

Figure 2: Basic servo loop.

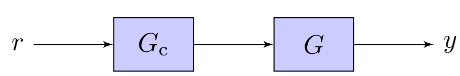

Figure 3: Feedforward controller.

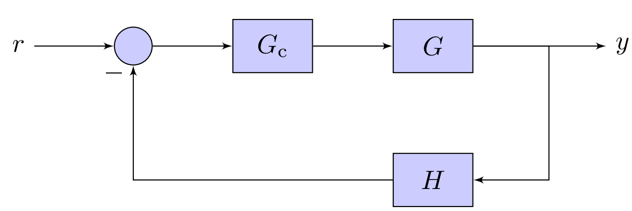

Figure 4: Basic servo loop with sensor dynamics.

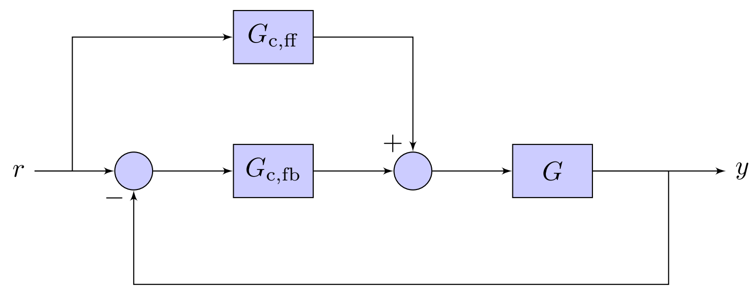

Figure 5: Basic servo loop with feedforward controller.

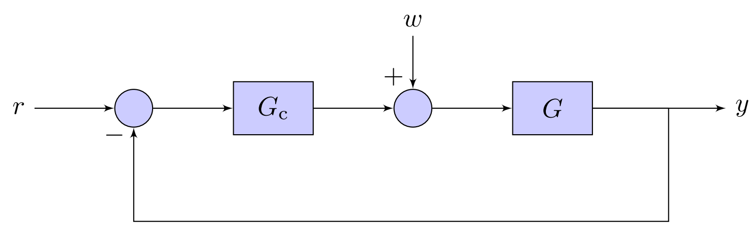

Figure 6: Basic servo loop with disturbance.

Hint: Follow the following steps.

1. Name input and output signals of each system.
2. For each subsystem, use the fact that the output is equal to the product of the transfer function and the input.
3. The signal coming out of the summing junction (cirlce) is the sum of all signals coming into the summing junction. Use appropriate signs.
4. Use all equations that you got to find the relation between r and y, and r and e.

The diagrams above illustrate several common modifications of the basic servo loop.

- A *feedforward path* allows the reference command to influence the plant input directly without passing through the feedback loop. Feedforward control can improve tracking performance when the plant dynamics are reasonably well-known.
- *Sensor dynamics* model the dynamics of the measurement device used to observe the output. Non-ideal sensors introduce additional dynamics into the feedback path that may affect stability and closed-loop performance.
- A *disturbance input* represents an external signal acting on the plant. Feedback control is often used to attenuate the effect of such disturbances on the output.

Problem 4. Consider a basic servo loop where the plant $G(s)$ is a

1. URB
2. DRB
3. UO
4. DO

and the controller $G_c(s)$ is 1) proportional controller, that is, $G_c(s) = K_p$ and 2) integral controller, that is, $G_c(s) = \frac{K_i}{s}$. For each controller and each plant type, derive the sensitivity transfer function $S(s)$. Then, if applicable, use the final value theorem to determine the asymptotic value of the error signal for a unit-step command input. If the final value theorem is not applicable, explain why.